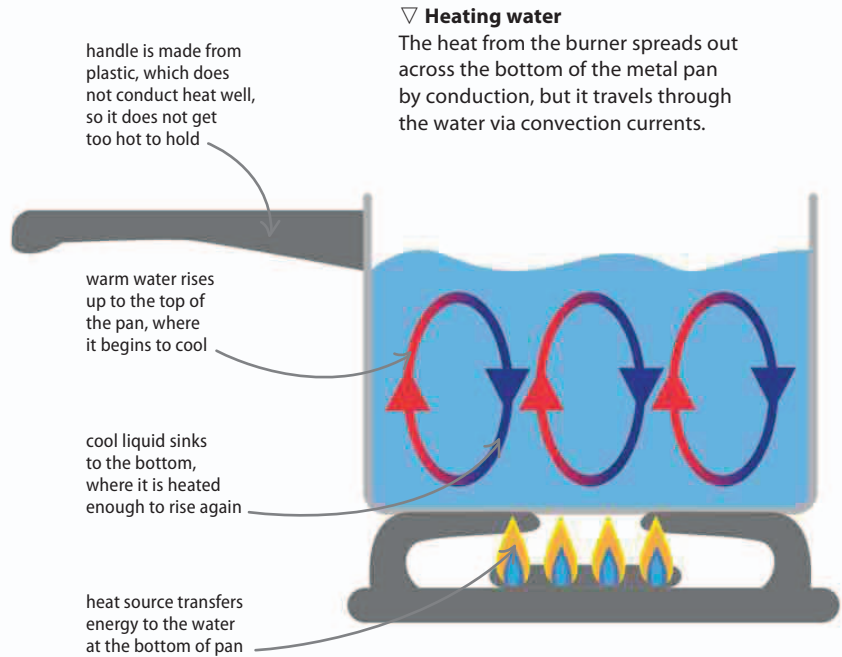


Convection currents

Heat moves through liquids or gases by a process called convection. This works on the basis that hot fluids rise upward while cooler ones sink. As a fluid receives energy its particles move faster and spread out. That results in it becoming less dense and rising upward through the cooler, denser fluid. The cooler fluid sinks, and fills the space left by the warmer fluid. This cool fluid is then exposed to the same heat source, and it is heated and rises up. As a result, heat is transferred around the fluid in a continuous convection current of rising and falling fluids.

Convection currents are responsible for the movement of **tectonic plates** on the Earth's crust.



▽ Heating water

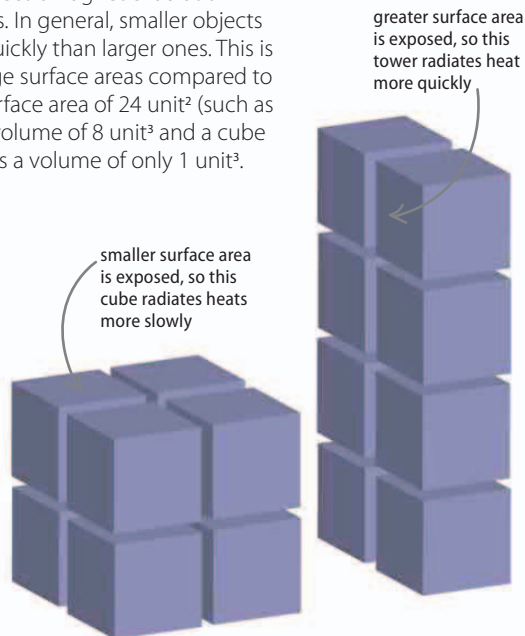
The heat from the burner spreads out across the bottom of the metal pan by conduction, but it travels through the water via convection currents.

Radiating heat

Heat can travel in the form of electromagnetic radiation—mainly infrared and microwaves. In general, smaller objects radiate away their heat more quickly than larger ones. This is because small objects have large surface areas compared to their volumes: a cube with a surface area of 24 unit² (such as the example below, left) has a volume of 8 unit³ and a cube with a surface area of 6 unit² has a volume of only 1 unit³.

▷ Comparing surface areas

This cube has the same volume (8 unit³) as the tower to the right, but has a smaller surface area (24 unit²). Therefore, less of its heat energy has access to the surface, so it radiates heat more slowly. The tower has a larger surface area (28 unit²) than the cube, so more of its heat energy has access to the surface, where it can radiate into space, making the object cool more quickly than the cube.



REAL WORLD

Saving heat

Animals that live in cold parts of the world are larger than their relatives that live in warmer places. For example, polar bears are much larger than the sun bears of southern Asia. The big polar animal loses precious heat more slowly than its tropical cousins because its large body gives it a small surface area to volume ratio.

